LESSON 6.9
APPLYING RATIO RELATIONSHIPS TO PERCENTAGES

LESSON INTRODUCTION

MA.6.AR.3.4 Apply ratio relationships to solve mathematical and real-world problems involving percentages using the relationship between two quantities.

LEARNING TARGETS

- I can make connections between ratios and percentages.
- I can use double line diagrams, tape diagrams, and ratio tables to solve problems involving percentages.

BENCHMARK COHERENCE

BUILDING ON

N/A

WORKING TOWARD

MA.7.AR.3.1 Apply previous understanding of percentages and ratios to solve multi-step real-world percent problems.

ESSENTIAL QUESTIONS

- How would you describe a percentage in terms of a ratio?
- How can you use double number lines, tape diagrams, and ratio tables to solve percent problems?

LESSON NARRATIVE

Students continue applying models of double number lines, tape diagrams, and ratio tables specifically in contexts of percentages and problems with percentages in the real-world. Students make sense of what percentage means through the lens of a ratio in order to approach problem-solving situations involving percentages similarly to previous lessons using a variety of models and strategies.

COMPONENT SUMMARY

6.9.1 WARM-UP *(~5 MIN.)*

Estimate the percentage – orienting students to content

6.9.2 REFRESHER *(10 MIN.)*

Review the concept of percent by representing percentages with a variety of models

6.9.3 GUIDED INSTRUCTION *(15-20 MIN.)*

Using a variety of models for problem-solving with percentages

6.9.4 GUIDED PRACTICE *(10 MIN.)*

Jigsaw practice for a variety of models

6.9.5 YOUR TURN! *(5-10 MIN.)*

Demonstrate understanding of using double number lines, ratio tables, and tape diagrams to solve percent problems

6.9.6 WRAP-UP *(~5 MIN.)*

Students reflect on their mastery of the lesson's learning targets

RESOURCES

GLOSSARY

Key Terms:

- Percent

Additional Terms: N/A

MATERIALS

- (Optional) Colored pencils
- (Optional) Graph paper

ADDITIONAL PREPARATION

- N/A

TEACHER TIPS

COMMON MISCONCEPTIONS

- Students may not make the same number of partitions when creating double number lines and tape diagrams (incorrect labeling or scaling).
- Students may not recognize that part-to-whole ratios connect to percentages rather than part-to-part ratios.

THINGS TO CONSIDER

- As an additional support, have students check that they have the same number of partitions in their tape diagram and double number lines.

LITERACY AND LANGUAGE ROUTINES

Think-Pair-Share

Consider having students engage in the 6.9.4 Guided Practice using the Think-Pair-Share routine. Students should independently solve the problems before discussing how they solved the problems and the answer they obtained with a partner. Encourage students to talk about their process more than their product, making note and surfacing any difference and variety in problem-solving techniques.

MATHEMATICAL THINKING AND REASONING (MTR) STANDARD

MA.K12.MTR.2.1 Multiple Representations

Throughout the lesson, students see ratios and percentages represented in multiple ways. Facilitate discussion to highlight a variety of strategies and representations, particularly noting differences in approach and strategy as students complete 6.9.4 Guided Practice.

DIFFERENTIATE INSTRUCTION

6.9.2 Refresher provides a visual entry point for students to visualize percentages. Consider providing students with graph paper and colored pencils so they can color a visual model of the percentages.

Consider allowing students to discuss their solution strategies and answers verbally with a partner for the 6.9.4 Guided Practice and 6.9.5 Your Turn! components. Discussing the problems will help develop student understanding and confidence.

6.9.1 WARM-UP: ESTIMATE THE PERCENTAGE

Component Overview: *Students estimate the percentage of thumb tacks that are blue from an image of a bunch of red and blue thumb tacks.*

TEACHER MOVES

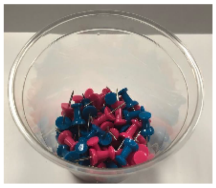
Turn and Talk

Consider having students participate in a “Turn and Talk” routine before answering question 2.

How can you use your estimates from question 1 to help you estimate for question 2?

6.9.1 Warm-Up: Estimate the Percentage

A photo of blue and pink thumb tacks is shown.



1. Based on what is shown in the picture, predict the percentage of thumb tacks that are blue by completing each statement. *Answers will vary.*

   I know for sure that more than __30__% of the thumb tacks are blue.

   I know for sure that less than __90__% of the thumb tacks are blue.

2. My best estimate is that __45__% of the thumb tacks are blue.

6.9.2 Refresher: Percentages

A ***percentage*** is a ratio expressed as a fraction of 100.

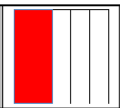
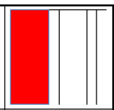
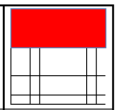
Percent (%) means "per 100," "for every 100," or "out of 100."

Work with your partner to complete the following.

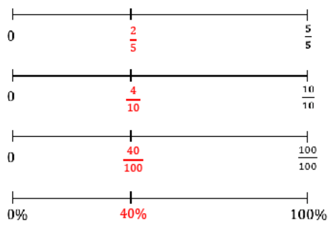
1. **Each of the large squares shown are the same size.**

   a. **Shade $\frac{2}{5}$ of each model.**

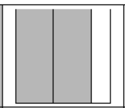
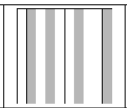
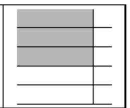
   b. **Then, express the shaded part of each model as a fraction and as a percentage.**

Model			
Fraction	$\frac{2}{5}$	$\frac{4}{10}$	$\frac{40}{100}$
Percentage	$\frac{2}{5} \times 100$ $= 40\%$	$\frac{4}{10} \times 100$ $= 40\%$	$\frac{40}{100} \times 100$ $= 40\%$

   c. **Label the identified tick marks on each number line below to match the models.**



2. **Express the shaded part of each model as a fraction and as a percentage.**

Model			
Fraction	$\frac{4}{5}$	$\frac{4}{10}$	$\frac{12}{25}$
Percentage	$\frac{4}{5} \times 100$ $= 80\%$	$\frac{4}{10} \times 100$ $= 40\%$	$\frac{12}{25} \times 100$ $= 48\%$

3. **Complete the statements.**
   Percentages represent [o part-to-part / o **part-to-whole**] **relationships where each percentage is a part per 100. In this case, 100% is** [o only part of the total. / o **the total, or whole, amount.**]

6.9.2 REFRESHER: PERCENTAGES

Component Overview: *Students review the concept of percent by representing percentages with area models, number lines, fractions, and percentages. Read the meaning of percent out loud for the class, then have students work in partners to represent percentages in different ways.*

DIFFERENTIATE INSTRUCTION

Consider providing students with graph paper and colored pencils so they can color a visual model of the percentages or encouraging students to model percentages by folding a piece of paper to provide a kinesthetic entry point.

TEACHER MOVES

Compare and Connect

After students have completed the table, consider engaging students in a "Compare and Connect" routine to establish the connections between the multiple ways to represent percentages. To facilitate conceptual understanding ask questions like:

- How does the model represent $\frac{2}{5}$?
- Why do you need to multiply $\frac{2}{5}$ by 100 to find the percent?
- What do you notice about all of the models and the fractions they represent?

6.9.3 GUIDED INSTRUCTION: USING RATIO MODELS TO REPRESENT PERCENTAGES

Component Overview: *This Guided Instruction begins with students using a double number line representation to find the percentage of water in a pot at different intervals. Students then connect the double number line to a table representation and explain how to relate to one another. Students then explore how to represent percentages using tape diagrams, using them to determine the number of students in each grade based on their percentage on the diagram.*

DIFFERENTIATE INSTRUCTION

MA.K12.MTR.2.1: Multiple Representations

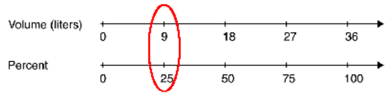
In this component, students see ratios and percentages represented in multiple ways. They are represented using tables, double number lines, and tape diagrams. Students draw connections among these representations throughout this component. These representations help students develop a conceptual understanding of percentages and how they are related to rates and ratios.

Consider incorporating questions that position students to evaluate which representation makes the most sense to them in each unique context.

6.9.3 Guided Instruction: Using Ratio Models to Represent Percentages

Double number lines, tape diagrams, and ratio tables can be used to model percentages similarly to other ratios. These can be helpful for determining equivalent ratios and solving problems involving percentages.

1. **Consider a cooking pot that can hold 36 liters of water. The double number line diagram models this situation.**



   a. **What percentage of the pot is filled when it contains 9 liters of water?**
   25%

   b. **Circle where the answer to the previous question is represented on the number line diagram.**

   c. **How many parts were the totals on each number line, 36 liters and 100%, split into?**
   4 parts

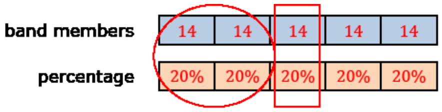
   d. **An equivalent ratio table can also be used to model similar thinking. Determine the factor that was multiplied by 36 to get 9, and use it to complete the blanks below.**

	Volume (liters)	Percentage	
$\times \frac{1}{4}$	36	100	$\times \frac{1}{4}$
	9	25	

   e. **Describe how the factor used to find the equivalent ratio in the table relates to how the number lines were divided in the number line diagram.**
   The factor used is to find the equivalent ratio is $\frac{1}{4}$. This relates to how the number lines were divided because each of them was divided into fourths.

Tape diagrams can also be used to represent problems involving percentages.

2. At Millennium Middle School, there are 70 students in the school band. Of the band members, 40% are sixth graders, 20% are seventh graders, and the rest are eighth graders.
The tape diagrams shown represent the situation.



a. Label each small rectangle in the tape diagram of the band members with the number of students it represents.

b. Label each small rectangle in the tape diagram of the percentage of band members with the value it represents.

c. How many band members are sixth graders?
28 band members are 6th graders.

d. Circle the part of the tape diagram that represents these students.

e. How many band members are seventh graders?
14 band members are 7th graders.

f. Draw a rectangle around the part of the diagram that represents these students.

g. What percentage of the band members are eighth graders?
40% are 8th graders.

h. How many band members are eighth graders?
28 band members are 8th graders.

6.9.3 GUIDED INSTRUCTION: USING RATIO MODELS TO REPRESENT PERCENTAGES (CONTINUED)

TEACHER MOVES

This Guided Instruction is intentionally scaffolded so students can draw connections between multiple visual representations and percentages.

Guide students through questions 2a and 2b as they determine how to use the tape diagrams to represent percentages of each grade in the school band.

Then, have students work in partners or small groups to use their tape diagrams to answer the questions. Circulate to observe how students are using their representations to make sense of each percentage.

TEACHER MOVES

Students might have trouble determining the number of students in each grade, especially because the total number of students is mentioned at the beginning of this component and not connected directly to the tape diagram.

Encourage students to label their tape diagrams to show the total number of students and ask probing questions to guide them through the process of determining the students in each grade.

- Which section of the tape diagram represents the number of students in 6th/7th/8th grade? How can you use this to find the total?
- How can you determine the number of students represented in each section of the tape diagram? Label each section if it will help keep track.

6.9.4 GUIDED PRACTICE: USING RATIO MODELS TO REPRESENT PERCENTAGES

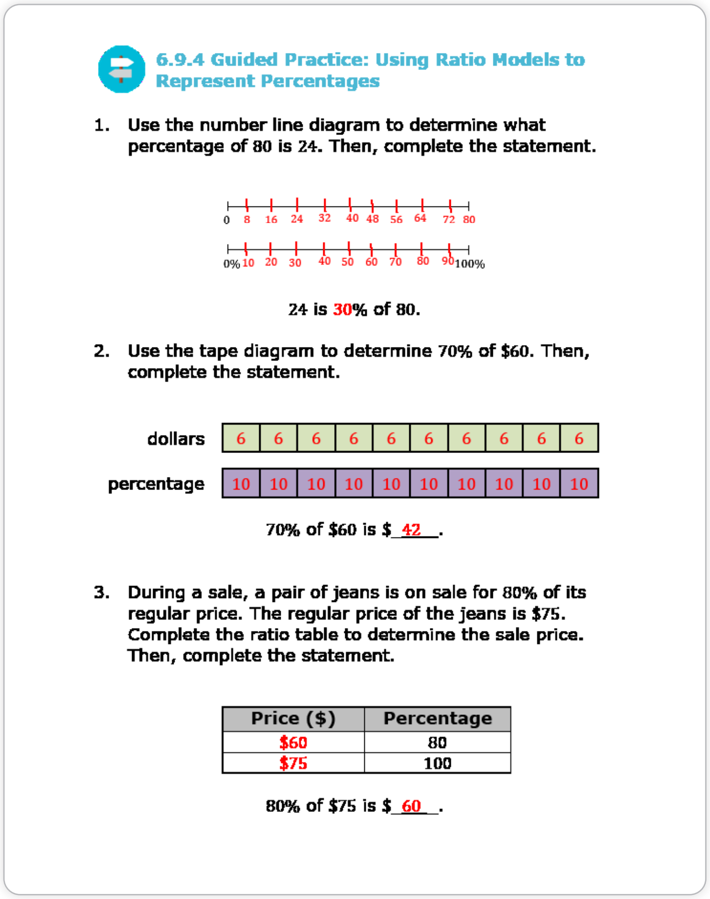
Component Overview: *Students practice using double number lines, ratio tables, and tape diagrams to solve percent problems.*

TEACHER MOVES

MA.K12.MTR.2.1: Multiple Representations

Notice the varying representations found in questions 1-3. When students have completed all three questions, consider prompting students to make connections between those representations and their effectiveness in problem-solving.

- Why was a number line used in question 1? Would a tape diagram or a table have been more effective? Why or why not? (answers will vary, encourage creativity and student choice)
- What about the tape diagram in question 2 or the table in question 3? Did those representations make the most sense in those contexts? Did you use something else? Why or why not?



6.9.4 Guided Practice: Using Ratio Models to Represent Percentages

1. Use the number line diagram to determine what percentage of 80 is 24. Then, complete the statement.

24 is 30% of 80.

2. Use the tape diagram to determine 70% of $60. Then, complete the statement.

dollars	6	6	6	6	6	6	6	6	6	6
percentage	10	10	10	10	10	10	10	10	10	10

70% of $60 is $_42__.

3. During a sale, a pair of jeans is on sale for 80% of its regular price. The regular price of the jeans is $75. Complete the ratio table to determine the sale price. Then, complete the statement.

Price ($)	Percentage
$60	80
$75	100

80% of $75 is $_60__.

6.9.5 Your Turn!

1. Use the models to find the values that complete each statement. Show your work.
For the last row, use the model of your choice.

Statement	Model
20% of 60 is 12.	0, 12, 24, 36, 48, 60 0%, 20%, 40%, 60%, 80%, 100%
48% of 25 is 12.	Value: 25, 12 / Percentage: 100, 48
40% of 70 is 28.	value: 7, 7, 7, 7, 7, 7, 7, 7, 7, 7 %: 10, 10, 10, 10, 10, 10, 10, 10, 10, 10
16% of 125 is 20.	Value: 125, 20 / Percentage: 100, 16 *Answers will vary.*

6.9.5 YOUR TURN!

Component Overview: *Students work independently to demonstrate their understanding of using double number lines, ratio tables, and tape diagrams to solve percent problems.*

DIFFERENTIATE INSTRUCTION

This component allows students to use the model of their choice to determine the last row in the table, allowing for multiple entry points to demonstrate understanding and learning.

Consider how student responses in this component could inform future small-group planning.

6.9.6 Wrap-Up: Reflection

Draw an **X** on each line to indicate where your current learning is related to each target.

1. I can make connections between ratios and percentages.

I need help. I can't get started. — I'm getting there, but I need more practice. — I understand this, and I feel confident.

Answers will vary.

2. I can use double-line diagrams, tape diagrams, and ratio tables to solve problems involving percentages.

I need help. I can't get started. — I'm getting there, but I need more practice. — I understand this, and I feel confident.

Answers will vary.

3. Write one question you still have about solving problems involving percentages after today's lesson.
Answers will vary.

6.9.6 WRAP-UP: REFLECTION

Component Overview: *Students reflect on their mastery of the lesson's learning targets.*

TEACHER MOVES

Consider using the following prompts as an additional data point for more accurate reflection:

- Which questions from the lesson involved making connections between ratios and percentages?
- Code them and think about how confident you felt answering these questions.
- Which questions from the lesson involved using double number lines, tape diagrams, and ratio tables to solve problems involving percentages?
- Code them and think about how confident you felt answering these questions.

Coding can include Mark-Up Texts like √ for confidence, ? for questions or help needed, or ± for getting there but need more practice.

